Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

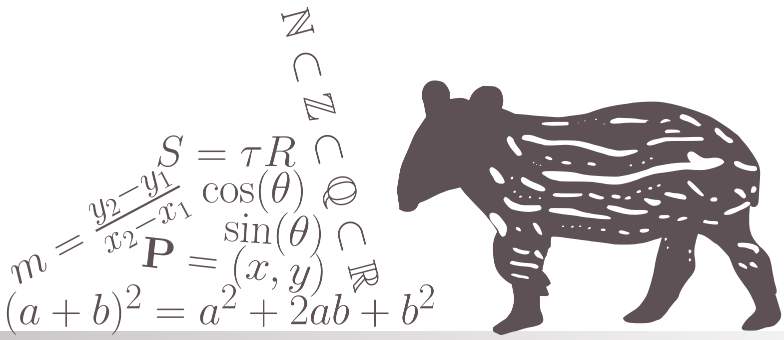
[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA



1.1 VECTORS AND VECTOR SPACES

1.1.1 Motivation

Motivating Problem 1.1 Bouncing Off a Wall

Imagine we are writing a 3D computer game in which an elastic ball bounces off a wall. The wall can be oriented in any possible direction in space, not necessarily aligned with the usual up/down, left/right or front/back directions. To correctly simulate this collision we need to know

1. when and where will the ball hit the wall, and
2. what would be the velocity of the ball immediately following the collision.

The goal of this motivating problem is to do so by using the parameters we already know:

1. the current position and velocity of the ball, and
2. the orientation and position of the wall.

To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3-dimensions, to understand ideas like points, lines and planes in 3D and how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.

Note: for simplicity (since this is not really about solving 3D physics) we'll assume that the ball is point-like, meaning that we don't need to care about its radius when doing these calculation.

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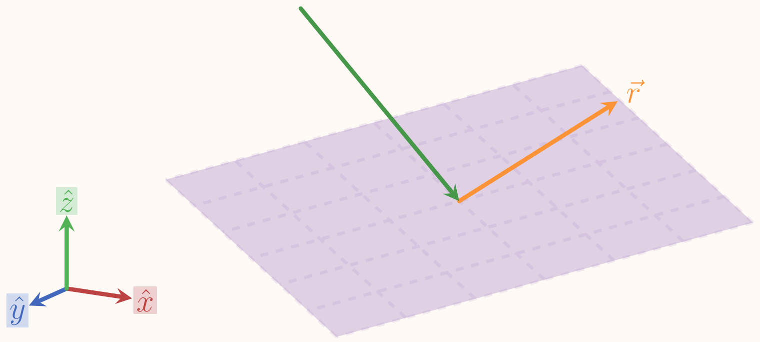

1.1.2 Geomteric Properties of Vectors

1.1.3 Geomteric Objects

1.1.4 Vector Spaces

1.1.5 Other Common Coordinate Systems in 2- and 3-Dimensions

1.1.6 Projections and Rejections

1.1.7 Solving the Motivating Problems

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